

Test your knowledge : Class_Quiz

Quiz . 30 Total Points

😊 Great job! Keep pushing for that perfect score! 🚀🌟

Your marks : 28 / 30

1. What is Apache Airflow?

- ☐ A distributed streaming platform.
- ☐ A high-performance data warehouse.
- ☒ A platform to programmatically author, schedule, and monitor workflows.
- ☐ A machine learning library.

✓ Correct!

2. In Airflow, what is a "DAG"?

- ☐ A data processing algorithm.
- ☒ A directed acyclic graph representing a workflow.
- ☐ A method to store data.
- ☐ A security protocol.

✓ Correct!

3. What does the term "operator" in Airflow represent?

- ☐ A mathematical computation.
- ☐ A type of error.
- ☒ A class that performs a specific function.
- ☐ An administrator user.

✓ Correct!

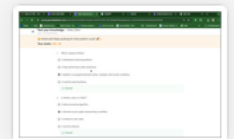
4. What is the purpose of the "airflow scheduler" command?

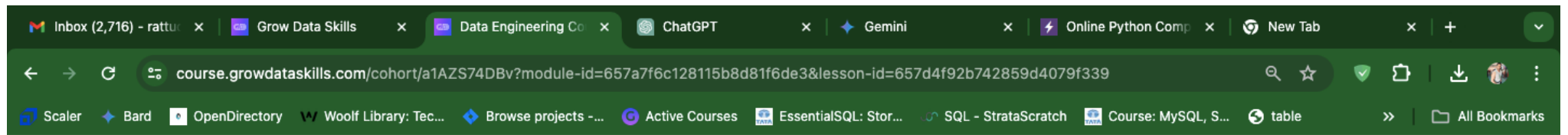
- ☐ It initializes the Airflow database.
- ☒ It schedules and monitors the execution of tasks.
- ☐ It deletes unused data.
- ☐ It creates a new DAG.

✓ Correct!

5. What is XCom in Airflow?

- ☐ An external communication system.
- ☒ A method for tasks to exchange messages.





5. What is XCom in Airflow?

- ☐ An external communication system.
- ☒ A method for tasks to exchange messages.
- ☐ A special type of operator.
- ☐ An error handling mechanism.

✓ Correct!

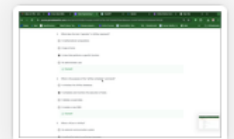
6. What does the Airflow "Backfill" command do?

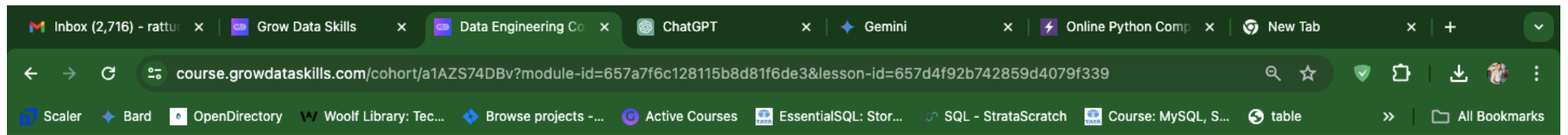
- ☐ It fills up the database with dummy data.
- ☒ It runs tasks for previous DAG runs for a specified date range.
- ☐ It repairs broken DAGs.
- ☐ It creates a backup of the current DAGs.

✓ Correct!

7. What command is used to start the Airflow web server?

- ☐ airflow web
- ☐ airflow start





7. What command is used to start the Airflow web server?

- ☐ airflow web
- ☐ airflow start
- ☒ airflow webserver
- ☐ airflow init

✓ Correct!

8. What is the function of "airflow initdb" command?

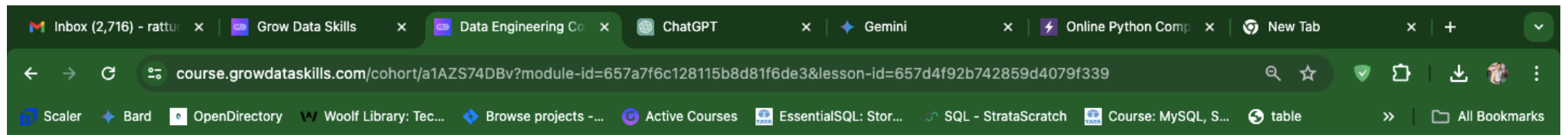
- ☒ Initialize the database.
- ☐ Delete the database.
- ☐ Repair the database.
- ☐ Back up the database.

✓ Correct!

9. Which of the following is a common use case for Airflow?

- ☒ Data extraction.
- ☐ Web server hosting.





9. Which of the following is a common use case for Airflow?

- ☒ Data extraction.
- ☐ Web server hosting.
- ☐ Image processing.
- ☐ Virtual reality rendering.

✓ Correct!

10. What is the main programming language used in Apache Airflow?

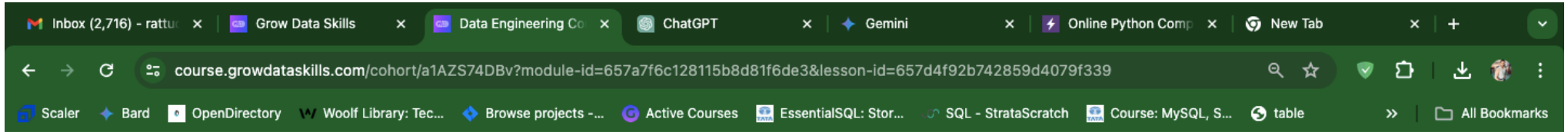
- ☒ Python
- ☐ Java
- ☐ C++
- ☐ Ruby

✓ Correct!

11. In Airflow, what is a task instance?

- ☐ An instance of an operator.
- ☐ A running DAG.





11. In Airflow, what is a task instance?

- ☐ An instance of an operator.
- ☐ A running DAG.
- ☒ An instance of a specific task in a DAG run.
- ☐ A snapshot of the database.

✓ Correct!

12. Which of the following is not a type of operator in Airflow?

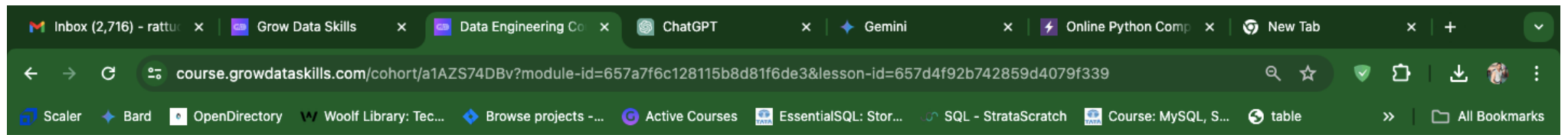
- ☐ BashOperator
- ☐ PythonOperator
- ☒ JavaOperator
- ☐ HttpOperator

✓ Correct!

13. What is the role of the Airflow executor?

- ☐ Scheduling tasks.
- ☒ Executing tasks.





13. What is the role of the Airflow executor?

- ☐ Scheduling tasks.
- ☒ Executing tasks.
- ☐ Defining tasks.
- ☐ Monitoring tasks.

✓ Correct!

14. Which command is used to check the status of Airflow tasks?

- ☐ airflow status
- ☐ airflow check
- ☐ airflow list_dags
- ☒ airflow task_state

✓ Correct!

15. What is the base operator in Airflow from which all other operators are derived?

- ☒ BaseOperator
- ☐ BasicOperator

15. What is the base operator in Airflow from which all other operators are derived?

- ☒ BaseOperator
- ☐ BasicOperator
- ☐ AirflowOperator
- ☐ MainOperator

✓ Correct!

16. How can you ensure a task runs after another in Airflow?

- ☐ Use the "depends_on_past" argument
- ☒ Use the "follows" argument.
- ☐ Use the "priority_weight" argument.
- ☐ Use the "run_after" argument.

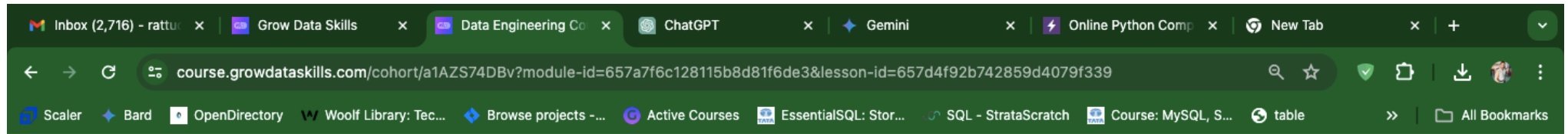
✗ Incorrect!

Use the "depends_on_past" argument is the correct answer

17. Which Airflow component shows the current state of all tasks?

- ☐ Airflow Scheduler.





17. Which Airflow component shows the current state of all tasks?

- ☐ Airflow Scheduler.
- ☒ Airflow Webserver.
- ☐ Airflow Executor.
- ☐ Airflow Worker.

✓ Correct!

18. How can a failed task be retried in Airflow?

- ☒ By using the "retries" parameter in the task.
- ☐ By manually restarting the task.
- ☐ By running the "airflow retry" command.
- ☐ Tasks cannot be retried in Airflow.

✓ Correct!

19. What does "airflow test" command do?

- ☐ It runs unit tests for Airflow.
- ☒ It tests a single task instance.



19. What does "airflow test" command do?

- ☐ It runs unit tests for Airflow.
- ☒ It tests a single task instance.
- ☐ It checks the validity of the DAGs.
- ☐ It checks the Airflow configuration.

✓ Correct!

20. What is the function of a sensor in Airflow?

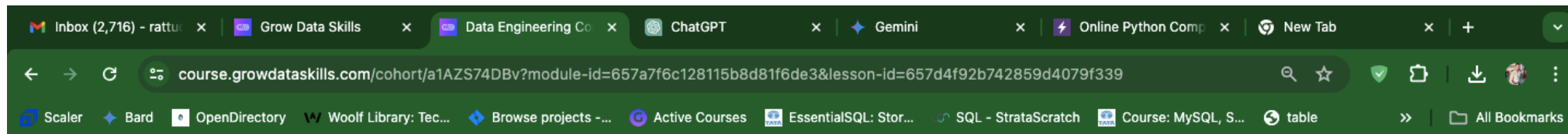
- ☐ It senses changes in the database.
- ☒ It is used for task synchronization.
- ☐ It measures the execution time of tasks.
- ☐ It senses changes in the Airflow configuration.

✓ Correct!

21. What does the "pool" concept in Airflow do?

- ☐ It groups similar tasks.
- ☒ It limits the parallel execution of tasks.





21. What does the "pool" concept in Airflow do?

- ☐ It groups similar tasks.
- ☒ It limits the parallel execution of tasks.
- ☐ It handles task dependencies.
- ☐ It schedules tasks.

✓ Correct!

22. What is a "Hook" in Airflow?

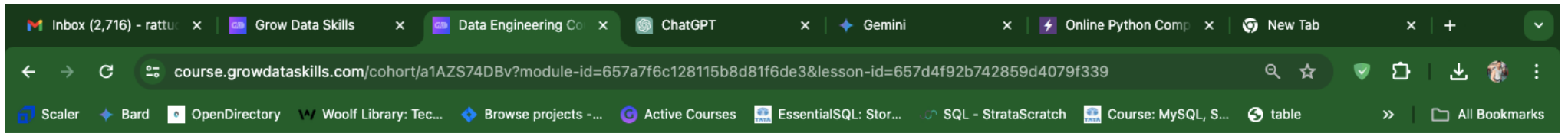
- ☒ A mechanism to connect to external systems.
- ☐ A type of sensor.
- ☐ A function to handle errors.
- ☐ A type of operator.

✓ Correct!

23. How are tasks organized in a DAG in Airflow?

- ☐ Randomly
- ☐ Alphabetically





23. How are tasks organized in a DAG in Airflow?

- ☐ Randomly
- ☐ Alphabetically
- ☐ Chronologically
- ☒ In a user-defined order.

✓ Correct!

24. Which of the following is not a state of a task in Airflow?

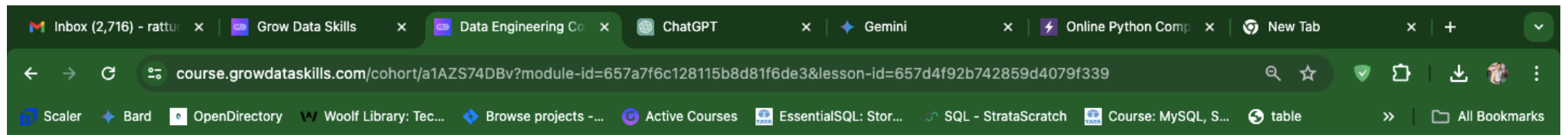
- ☐ Running
- ☐ Success
- ☐ Error
- ☒ Paused

✓ Correct!

25. What language is the configuration file of Airflow written in?

- ☐ JSON
- ☐ XML





25. What language is the configuration file of Airflow written in?

- ☐ JSON
- ☐ XML
- ☐ YAML
- ☒ Python

✓ Correct!

26. Which of the following cannot be defined in the Airflow configuration file?

- ☐ Executor
- ☐ Database
- ☐ Scheduler
- ☒ DAGs

✓ Correct!

27. Which Airflow component executes a task instance?

- ☐ Scheduler
- ☒ Executor



27. Which Airflow component executes a task instance?

- ☐ Scheduler
- ☒ Executor
- ☐ Webserver
- ☐ Worker

✗ **Incorrect!**
Worker is the correct answer

28. Which one of the following is not an executor in Airflow?

- ☐ SequentialExecutor
- ☐ LocalExecutor
- ☐ CeleryExecutor
- ☒ FastExecutor

✓ **Correct!**

29. What does the "trigger_rule" parameter in a task do in Airflow?

- ☒ It defines the condition under which the task is triggered.



29. What does the "trigger_rule" parameter in a task do in Airflow?

- ☒ It defines the condition under which the task is triggered.
- ☐ It defines the order of task execution.
- ☐ It defines the retry mechanism for the task.
- ☐ It defines the priority of the task.

✓ Correct!

30. What is "airflow upgradedb" command used for?

- ☐ It upgrades Airflow to the latest version.
- ☒ It upgrades the database schema to match the Airflow version.
- ☐ It upgrades the Airflow configuration.
- ☐ It upgrades the Airflow plugins.

✓ Correct!

Submit

